

User Document: Project 4

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INTRODUCTION

This system is an enhanced version of a previous project that efficiently retrieves and displays geographical data from a large dataset. The original system read data from a CSV file and displayed a table showing the easternmost, westernmost, northernmost, and southernmost zip codes for each state. The CSV file could be in any order, and the system worked effectively even with randomly sorted data.

The current system builds on this foundation by providing additional features and improvements. It introduces a file converter that transforms the original CSV file into a custom Zipcode Record (ZCR) file. This ZCR file maintains the integrity of the original data while adding additional information to enhance the user's understanding and control of the dataset. The ZCR file includes a fixed-size header that provides high-level information about the data. Each header field is clearly labeled and separated by semicolons for easy reading.

The system also introduces a new buffer system designed to handle the ZCR file format which will improve usage from the data file. This new system has been thoroughly tested with the original program to ensure compatibility and performance.

One of the key enhancements in this iteration is the implementation of a primary key index based on the zipcode field value. This index is stored in a separate Index (IDX) file and can be loaded into memory for fast data processing, reducing the need for expensive disk accesses. The IDX file allows the system to quickly locate specific records by zipcode.

The system is designed to be user-friendly and can be operated from the command line. It allows users to input the name of the CSV or ZCR file to be processed and provides functionality to convert a CSV file into a custom ZCR file. The system can locate a record based on user input or based on command-line input with the -Z flag. If the record is found, it is displayed; if not, a message is displayed stating that the record is not present. The system can access multiple records in a single execution.

This user document provides a comprehensive guide to using the system. It includes detailed descriptions of the system's features and functionalities, step-by-step instructions for performing common tasks, and solutions to potential issues that users may encounter. The document is organized into sections for easy navigation and understanding. It is written in clear and concise language, with diagrams and examples where necessary to aid understanding. The document is regularly updated to reflect the latest changes and improvements to the system. It is intended for end-users who want to understand how to use the system effectively. Whether you are a new user learning to use the system for the first time or an experienced user looking for information on advanced features, this document is designed to help you get the most out of the system.

SAMPLE FILES WITH ZIPCODE RECORDS

Sample 1: .CSV with standard organization and sort

1	Zip Code,Place Name,State,County,Lat,Long
2	501,Holtsville,NY,Suffolk,40.8154,-73.0451
3	544,Holtsville,NY,Suffolk,40.8154,-73.0451
4	1001,Agawam,MA,Hampden,42.0702,-72.6227
5	1002,Amherst,MA,Hampshire,42.3671,-72.4646
6	1003,Amherst,MA,Hampshire,42.3919,-72.5248
7	1004,Amherst,MA,Hampshire,42.3845,-72.5132
8	1005,Barre,MA,Worcester,42.4097,-72.1084
9	1007,Belchertown,MA,Hampshire,42.2751,-72.411
10	1008,Blandford,MA,Hampden,42.1829,-72.9361
11	1009,Bondsville,MA,Hampden,42.2061,-72.3405
12	1010,Brimfield,MA,Hampden,42.1165,-72.1885
13	1011,Chester,MA,Hampden,42.2794,-72.9888
14	1012,Chesterfield,MA,Hampshire,42.3923,-72.8256
15	1013,Chicopee,MA,Hampden,42.1487,-72.6079
16	1014,Chicopee,MA,Hampden,42.1707,-72.6048
17	1020,Chicopee,MA,Hampden,42.1764,-72.5761
18	1021,Chicopee,MA,Hampden,42.1707,-72.6048
19	1022,Chicopee,MA,Hampden,42.1934,-72.5544
20	1026,Cummington,MA,Hampshire,42.4633,-72.9202
21	1027,Easthampton,MA,Hampshire,42.2668,-72.669
22	1028,East Longmeadow,MA,Hampden,42.0672,-72.5056
23	1029,East Otis,MA,Berkshire,42.1909,-73.0517
24	1030,Feeding Hills,MA,Hampden,42.0718,-72.6751
25	1031,Gilbertville,MA,Worcester,42.3322,-72.1986

Sample 2: .ZCR with reorganized columns and random rows.

1	File Structure Type: ZCR;
2	Version: 1.0;
3	Header Record Size: 450;
4	Record Size Integer Bytes: -1;
5	Size Format Type: ASCII;
6	Index File Name: us_postal_codes_rr_state.idx;
7	Index File Schema: Key, Offset;
8	Record Count: 40933;
9	Fields Per Record: 6;
10	Field 1: State, String;
11	Field 2: Zip Code, Int;
12	Field 3: Place Name, String;
13	Field 4: County, String;
14	Field 5: Lat, Double;
15	Field 6: Long, Double;
16	Primary Key: Zip Code;
17	43 HI,96737,Ocean View,Hawaii,19.1002,-155.726
18	45 OR,97134,Oceanside,Tillamook,45.4549,-123.965
19	55 MO,63701,Cape Girardeau,Cape Girardeau,37.3169,-89.5459
20	34 NM,88250,Hope,Eddy,32.8156,-104.73
21	41 MO,64444,Edgerton,Platte,39.4742,-94.6352
22	42 PA,15450,La Belle,Fayette,39.9983,-79.9754
23	47 CA,94278,Sacramento,Sacramento,38.5816,-121.494
24	43 OR,97875,Stanfield,Umatilla,45.786,-119.212
25	42 NE,68113,Offutt Afb,Sarpy,41.2563,-95.9404
26	43 IL,61536,Hanna City,Peoria,40.6798,-89.7952
27	42 HI,96837,Honolulu,Honolulu,21.315,-157.863
28	51 CA,93465,Templeton,San Luis Obispo,35.5551,-120.711
29	40 CA,92807,Anaheim,Orange,33.8544,-117.786
30	42 SC,29936,Ridgeland,Jasper,32.4807,-80.9804
31	37 OH,45646,Latham,Pike,39.0804,-83.3283
32	38 WA,98119,Seattle,King,47.6379,-122.364
33	44 PA,18660,Wapwallopen,Luzerne,41.068,-76.0857
34	39 OH,45769,Pomeroy,Meigs,39.0607,-82.0331
35	40 AR,72450,Paragould,Greene,36.06,-90.5251
36	38 NY,14058,Elba,Genesee,43.0897,-78.1704
37	36 PA,16023,Cabot,Butler,40.7924,-79.76
38	44 TN,38041,Henning,Lauderdale,35.6593,-89.6962
39	40 MS,38754,Isola,Humphreys,33.2471,-90.604
40	42 NY,12592,Wassaic,Dutchess,41.7759,-73.5544
41	39 OK,74761,Swink,Choctaw,34.0179,-95.2022
42	47 FL,33862,Lake Placid,Highlands,27.2931,-81.3629
43	43 MN,55329,Eden Valley,Meeker,45.302,-94.6039
44	39 NC,28692,Vilas,Watauga,36.2574,-81.7652
45	39 MN,56563,Moorhead,Clay,46.8739,-96.7695
46	38 CA,95913,Artois,Glenn,39.6197,-122.193
47	45 IL,62794,Springfield,Sangamon,39.7495,-89.606
48	39 ME,4022,Denmark,Oxford,43.9755,-70.7924
49	37 AR,71748,Ivan,Dallas,33.9118,-92.4246
50	50 TX,76908,Goodfellow Afb,Tom Green,31.4321,-100.402

HOW TO

Overview: The system consists of 4 subprograms. Each subprogram provides a functionality for accessing Zipcode records from different file types. All subprograms can be accessed at:

`/workspaces/csci331-spring-24-project4-dagron27` on GitHub.

A program can be called by using:

`./<insert subprogram>`

In addition, proj4 and its variants can accept command line flag “-Z”:

`./proj4_norm -Z544`

Each subprogram will be detailed below. The program name, a brief summary, input type, output, instruction, and troubleshooting are provided. Screenshots are provided for easy visibility.

Subprogram: proj3_original

Summary: The program is designed to analyze geographical data from a CSV file and display extreme zip codes (easternmost, westernmost, northernmost, and southernmost) for each state.

Input: properly organized .csv file (row order does not matter, column order matters)

Output: ordered table listed by state providing eastmost, westmost, northmost, and southmost extrema

Examples

Example 1: us_postal_codes.csv

```
@dagron27 → /workspaces/csci331-spring-24-project4-dagron27 (main) $ ./proj3_original
Enter the name of the CSV file to be opened: us_postal_codes.csv
State      Easternmost  Westernmost  Northernmost  Southernmost
AA         -82.2493     -82.2493     33.0364       33.0364
AK         -176.6581    -130.0247    71.2346       51.8800
AL         -88.3865     -84.9972     34.9804       30.2460
AP         -155.7982    -85.8200     41.7206       19.1110
AR         -94.5783     -89.7981     36.4864       33.0163
AZ         -114.7822    -109.0537    36.9903       31.3373
CA         -124.2514    -114.1430    41.9404       32.5626
CO         -108.9799    -102.1415    40.9979       37.0028
CT         -73.6594     -71.8026     42.0320       41.0265
DC         -77.0860     -76.9376     38.9757       38.8262
DE         -75.7584     -75.0589     39.8188       38.4654
FL         -87.4539     -80.0388     30.9821       24.5552
FM         151.5031     151.5031     7.1383        7.1383
GA         -85.5171     -80.8509     34.9814       30.5224
HI         -159.9214    166.6419     22.2192       19.0668
IA         -96.5225     -90.1885     43.4899       40.4094
ID         -117.0002    -111.0722    48.9994       42.0288
IL         -91.4348     -87.5446     42.4906       37.0123
IN         -87.9172     -84.8161     41.7427       37.8858
KS         -102.0070    -94.6144     39.9845       37.0005
KY         -89.1947     -82.0784     39.1114       36.5422
LA         -94.0034     -89.2576     33.0005       29.1816
MA         -73.4566     -69.9620     42.8559       41.2639
MD         -79.4219     -75.0849     39.7187       37.9666
ME         -71.0112     -67.0074     47.4589       43.0921
MH         167.7365     171.2370     8.7860        7.1128
MI         -90.1588     -82.4249     47.4688       41.7361
MN         -97.2045     -89.6989     49.3453       43.5084
```

Example 2: us_postal_codes_ROWS_RANDOMIZED.csv

```
@dagron27 → /workspaces/csci331-spring-24-project4-dagron27 (main) $ ./proj3_original
Enter the name of the CSV file to be opened: us_postal_codes_ROWS_RANDOMIZED.csv
```

State	Easternmost	Westernmost	Northernmost	Southernmost
AA	-82.2493	-82.2493	33.0364	33.0364
AK	-176.6581	-130.0247	71.2346	51.8800
AL	-88.3865	-84.9972	34.9804	30.2460
AP	-155.7982	-85.8200	41.7206	19.1110
AR	-94.5783	-89.7981	36.4864	33.0163
AZ	-114.7822	-109.0537	36.9903	31.3373
CA	-124.2514	-114.1430	41.9404	32.5626
CO	-108.9799	-102.1415	40.9979	37.0028
CT	-73.6594	-71.8026	42.0320	41.0265
DC	-77.0860	-76.9376	38.9757	38.8262
DE	-75.7584	-75.0589	39.8188	38.4654
FL	-87.4539	-80.0388	30.9821	24.5552
FM	151.5031	151.5031	7.1383	7.1383
GA	-85.5171	-80.8509	34.9814	30.5224
HI	-159.9214	166.6419	22.2192	19.0668
IA	-96.5225	-90.1885	43.4899	40.4094
ID	-117.0002	-111.0722	48.9994	42.0288
IL	-91.4348	-87.5446	42.4906	37.0123
IN	-87.9172	-84.8161	41.7427	37.8858
KS	-102.0070	-94.6144	39.9845	37.0005
KY	-89.1947	-82.0784	39.1114	36.5422
LA	-94.0034	-89.2576	33.0005	29.1816
MA	-73.4566	-69.9620	42.8559	41.2639
MD	-79.4219	-75.0849	39.7187	37.9666
ME	-71.0112	-67.0074	47.4589	43.0921
MH	167.7365	171.2370	8.7860	7.1128
MI	-90.1588	-82.4249	47.4688	41.7361
MN	-97.2045	-89.6989	49.3453	43.5084

Troubleshooting

Example 1: .csv with incorrect column order

```
● @dagron27 → /workspaces/csci331-spring-24-project4-dagron27 (main) $ ./proj3_original
Enter the name of the CSV file to be opened: us_postal_codes_place.csv
State      Easternmost  Westernmost  Northernmost  Southernmost  _
```

Solution: Exam .csv to ensure it meets the standard format

Example 2: incorrect file format

```
● @dagron27 → /workspaces/csci331-spring-24-project4-dagron27 (main) $ ./proj3_original
Enter the name of the CSV file to be opened: us_postal_codes.zcr
State      Easternmost  Westernmost  Northernmost  Southernmost
○ @dagron27 → /workspaces/csci331-spring-24-project4-dagron27 (main) $ █
```

Solution: Ensure proper file format

Example 3: incorrect input / file does not exist

```
⊗ @dagron27 → /workspaces/csci331-spring-24-project4-dagron27 (main) $ ./proj3_original
Enter the name of the CSV file to be opened: sdfsfssf
Failed to open file: sdfsfssf
```

Solution: Verify input is correct or file exists

Subprogram: proj3_updated

Summary: The program is designed to analyze geographical data from a ZCR or CSV file and display extreme zip codes (easternmost, westernmost, northernmost, and southernmost) for each state.

Input: properly organized .zcr or .csv file (row order does not matter, column order matters)

Output: ordered table listed by state providing eastmost, westmost, northmost, and southmost extrema

Example 1: us_postal_codes.zcr

```

@dagron27 → /workspaces/csci331-spring-24-project4-dagron27 (main) $ ./proj3_updated
Enter the name of the ZCR file to be opened: us_postal_codes.zcr
State      Easternmost  Westernmost  Northernmost  Southernmost
AA          -82.2493     -82.2493     33.0364       33.0364
AK          -176.6580    -130.0250    71.2346       51.8800
AL          -88.3865     -84.9972     34.9804       30.2460
AP          -155.7980    -85.8200     41.7206       19.1110
AR          -94.5783     -89.7981     36.4864       33.0163
AZ          -114.7820    -109.0540    36.9903       31.3373
CA          -124.2510    -114.1430    41.9404       32.5626
CO          -108.9800    -102.1410    40.9979       37.0028
CT          -73.6594     -71.8026     42.0320       41.0265
DC          -77.0860     -76.9376     38.9757       38.8262
DE          -75.7584     -75.0589     39.8188       38.4654
FL          -87.4539     -80.0388     30.9821       24.5552
FM          151.5030     151.5030     7.1383        7.1383
GA          -85.5171     -80.8509     34.9814       30.5224
HI          -159.9210    166.6420     22.2192       19.0668
IA          -96.5225     -90.1885     43.4899       40.4094
ID          -117.0000    -111.0720    48.9994       42.0288
IL          -91.4348     -87.5446     42.4906       37.0123
IN          -87.9172     -84.8161     41.7427       37.8858
KS          -102.0070    -94.6144     39.9845       37.0005
KY          -89.1947     -82.0784     39.1114       36.5422
LA          -94.0034     -89.2576     33.0005       29.1816
MA          -73.4566     -69.9620     42.8559       41.2639
MD          -79.4219     -75.0849     39.7187       37.9666
ME          -71.0112     -67.0074     47.4589       43.0921
MH          167.7370     171.2370     8.7860        7.1128
MI          -90.1588     -82.4249     47.4688       41.7361
MN          -97.2045     -89.6989     49.3453       43.5084

```


Example 2: us_postal_codes_ROWS_RANDOMIZED.zcr

```

@dagron27 → /workspaces/csci331-spring-24-project4-dagron27 (main) $ ./proj3_updated
Enter the name of the ZCR file to be opened: us_postal_codes_ROWS_RANDOMIZED.zcr

```

State	Easternmost	Westernmost	Northernmost	Southernmost
AA	-82.2493	-82.2493	33.0364	33.0364
AK	-176.6580	-130.0250	71.2346	51.8800
AL	-88.3865	-84.9972	34.9804	30.2460
AP	-155.7980	-85.8200	41.7206	19.1110
AR	-94.5783	-89.7981	36.4864	33.0163
AZ	-114.7820	-109.0540	36.9903	31.3373
CA	-124.2510	-114.1430	41.9404	32.5626
CO	-108.9800	-102.1410	40.9979	37.0028
CT	-73.6594	-71.8026	42.0320	41.0265
DC	-77.0860	-76.9376	38.9757	38.8262
DE	-75.7584	-75.0589	39.8188	38.4654
FL	-87.4539	-80.0388	30.9821	24.5552
FM	151.5030	151.5030	7.1383	7.1383
GA	-85.5171	-80.8509	34.9814	30.5224
HI	-159.9210	166.6420	22.2192	19.0668
IA	-96.5225	-90.1885	43.4899	40.4094
ID	-117.0000	-111.0720	48.9994	42.0288
IL	-91.4348	-87.5446	42.4906	37.0123
IN	-87.9172	-84.8161	41.7427	37.8858
KS	-102.0070	-94.6144	39.9845	37.0005
KY	-89.1947	-82.0784	39.1114	36.5422
LA	-94.0034	-89.2576	33.0005	29.1816
MA	-73.4566	-69.9620	42.8559	41.2639
MD	-79.4219	-75.0849	39.7187	37.9666
ME	-71.0112	-67.0074	47.4589	43.0921
MH	167.7370	171.2370	8.7860	7.1128
MI	-90.1588	-82.4249	47.4688	41.7361
MN	-97.2045	-89.6989	49.3453	43.5084

Example 3: us_postal_codes.csv

```

• @dagron27 → /workspaces/csci331-spring-24-project4-dagron27 (main) $ ./proj3_updated
Enter the name of the ZCR file to be opened: us_postal_codes.csv
State      Easternmost  Westernmost  Northernmost  Southernmost
AA          -82.2493     -82.2493     33.0364       33.0364
AK          -176.6581    -130.0247    71.2346       51.8800
AL          -88.3865     -84.9972     34.9804       30.2460
AP          -155.7982    -85.8200     41.7206       19.1110
AR          -94.5783     -89.7981     36.4864       33.0163
AZ          -114.7822    -109.0537    36.9903       31.3373
CA          -124.2514    -114.1430    41.9404       32.5626
CO          -108.9799    -102.1415    40.9979       37.0028
CT          -73.6594     -71.8026     42.0320       41.0265
DC          -77.0860     -76.9376     38.9757       38.8262
DE          -75.7584     -75.0589     39.8188       38.4654
FL          -87.4539     -80.0388     30.9821       24.5552
FM          151.5031     151.5031     7.1383        7.1383
GA          -85.5171     -80.8509     34.9814       30.5224
HI          -159.9214    166.6419     22.2192       19.0668
IA          -96.5225     -90.1885     43.4899       40.4094
ID          -117.0002    -111.0722    48.9994       42.0288
IL          -91.4348     -87.5446     42.4906       37.0123
IN          -87.9172     -84.8161     41.7427       37.8858
KS          -102.0070    -94.6144     39.9845       37.0005
KY          -89.1947     -82.0784     39.1114       36.5422
LA          -94.0034     -89.2576     33.0005       29.1816
MA          -73.4566     -69.9620     42.8559       41.2639
MD          -79.4219     -75.0849     39.7187       37.9666
ME          -71.0112     -67.0074     47.4589       43.0921
MH          167.7365     171.2370     8.7860        7.1128
MI          -90.1588     -82.4249     47.4688       41.7361
MN          -97.2045     -89.6989     49.3453       43.5084

```

Troubleshooting

Example 1: .zcr with incorrect column order

```
• @dagron27 → /workspaces/csci331-spring-24-project4-dagron27 (main) $ ./proj3_updated
Enter the name of the ZCR file to be opened: us_postal_codes_place.zcr
Invalid argument: stoi
State      Easternmost  Westernmost  Northernmost  Southernmost  _
```

Solution: Exam .zcr to ensure it meets the standard format

Example 2: incorrect input / file does not exist

```
• @dagron27 → /workspaces/csci331-spring-24-project4-dagron27 (main) $ ./proj3_updated
Enter the name of the ZCR file to be opened: sfsdfsdf
File is not open. Cannot read from file.
State      Easternmost  Westernmost  Northernmost  Southernmost
```

Solution: Verify input is correct or file exists

Subprogram: converter

Summary: The program is designed to take a .csv containing a Zipcode record data set and convert it to a ZCR format.

Input: properly organized .csv file (row order does not matter, column order matters)

Output: a .zcr will be created or updated with the original file name but new extension, .zcr

Examples

Example:

Step 1: Choose a .csv file. We will use “test.csv” which contains a randomly sorted data set.

```
● @dagron27 → /workspaces/csci331-spring-24-project4-dagron27 (main) $ head -n 20 test3.csv
"Zip Code", "Place Name", State, County, Lat, Long
96737, Ocean View, HI, Hawaii, 19.1002, -155.7258
97134, Oceanside, OR, Tillamook, 45.4549, -123.965
63701, Cape Girardeau, MO, Cape Girardeau, 37.3169, -89.5459
88250, Hope, NM, Eddy, 32.8156, -104.7299
64444, Edgerton, MO, Platte, 39.4742, -94.6352
15450, La Belle, PA, Fayette, 39.9983, -79.9754
94278, Sacramento, CA, Sacramento, 38.5816, -121.4944
97875, Stanfield, OR, Umatilla, 45.786, -119.2116
68113, Offutt Afb, NE, Sarpy, 41.2563, -95.9404
61536, Hanna City, IL, Peoria, 40.6798, -89.7952
96837, Honolulu, HI, Honolulu, 21.315, -157.8633
93465, Templeton, CA, San Luis Obispo, 35.5551, -120.7107
92807, Anaheim, CA, Orange, 33.8544, -117.7858
29936, Ridgeland, SC, Jasper, 32.4807, -80.9804
45646, Latham, OH, Pike, 39.0804, -83.3283
98119, Seattle, WA, King, 47.6379, -122.3643
18660, Wapwallopen, PA, Luzerne, 41.068, -76.0857
45769, Pomeroy, OH, Meigs, 39.0607, -82.0331
72450, Paragould, AR, Greene, 36.06, -90.5251
```

Step 2: Run program “converter” and input the file name.

```

@dagron27 → /workspaces/csci331-spring-24-project4-dagron27 (main) $ ./converter
Enter the name of the CSV file: test3.csv
Program began...
Step 1: File < test3.csv > opened successfully
Step 2: File < test3.zcr > opened successfully
Step 3: Default header written to new .zcr
Step 4: Records from .cvs written to new .zcr
Step 5: Header updated for new .zcr
File conversion complete...

```

Step 3: View new .zcr file for accuracy.

```

@dagron27 → /workspaces/csci331-spring-24-project4-dagron27 (main) $ head -n 20 test3.zcr
File Structure Type: ZCR;
Version: 1.0;
Header Record Size: 450;
Record Size Integer Bytes: -1;
Size Format Type: ASCII;
Index File Name: primaryKey.idx;
Index File Schema: Key, Offset;
Record Count: 40933;
Fields Per Record: 6;
Field 1: Zip Code, Int;
Field 2: Place Name, String;
Field 3: State, String;
Field 4: County, String;
Field 5: Lat, Double;
Field 6: Long, Double;
Primary Key: Zip Code;
43 96737,Ocean View,HI,Hawaii,19.1002,-155.726
45 97134,Oceanside,OR,Tillamook,45.4549,-123.965
55 63701,Cape Girardeau,MO,Cape Girardeau,37.3169,-89.5459

```

Troubleshooting

Example 1: incorrect file format

```
⊗ @dagron27 → /workspaces/csci331-spring-24-project4-dagron27 (main) $ ./converter
Enter the name of the CSV file: test3.zcr
Program began...
Error: Input file is not a CSV file.
```

Solution: Ensure proper file format

Example 2: incorrect input / file does not exist

```
⊗ @dagron27 → /workspaces/csci331-spring-24-project4-dagron27 (main) $ ./converter
Enter the name of the CSV file: sffsfdsf
Program began...
terminate called after throwing an instance of 'std::out_of_range'
  what():  basic_string::substr: __pos (which is 18446744073709551615) > this->size() (which is 8)
Aborted (core dumped)
```

Solution: Verify input is correct or file exists

Example 3: incorrect column order

```
● @dagron27 → /workspaces/csci331-spring-24-project4-dagron27 (main) $ ./converter
Enter the name of the CSV file: us_postal_codes_place.csv
Program began...
Step 1: File < us_postal_codes_place.csv > opened successfully
Step 2: File < us_postal_codes_place.zcr > opened successfully
Step 3: Default header written to new .zcr
Step 4: Records from .csv written to new .zcr
Step 5: Header updated for new .zcr
File conversion complete...
```

```
File Structure Type: ZCR;
Version: 1.0;
Header Record Size: 450;
Record Size Integer Bytes: -1;
Size Format Type: ASCII;
Index File Name: primaryKey.idx;
Index File Schema: Key, Offset;
Record Count: 40933;
Fields Per Record: 6;
Field 1: Zip Code, Int;
Field 2: Place Name, String;
Field 3: State, String;
Field 4: County, String;
Field 5: Lat, Double;
Field 6: Long, Double;
Primary Key: Zip Code;
```

Solution: Verify file organization

Subprogram: proj4_norm (variations provided to utilize differently organized data sets)

Variations: proj4_norm_place, proj_rand, proj_rand_state

Summary: The program is designed to take a user input to print one Zipcode record, or command line inputs to print one or more Zipcode records.

Input: User input of valid zipcode. The -Z flag can be used from the command line.

Output: One or more Zipcode records will be printed to the console, if valid. If not valid, a message will display.

Example 1: user input of valid Zipcode record

```
● @dagron27 → /workspaces/csci331-spring-24-project4-dagron27 (main) $ ./proj4_norm
Please enter a zipcode: 544
Zip Code: 544
Place Name: Holtsville
State: NY
County: Suffolk
Latitude: 40.8154
Longitude: -73.0451
```

Example 2: command line -Z of valid Zipcode record

```
● @dagron27 → /workspaces/csci331-spring-24-project4-dagron27 (main) $ ./proj4_norm -Z544
Zip Code: 544
Place Name: Holtsville
State: NY
County: Suffolk
Latitude: 40.8154
Longitude: -73.0451
```

Example 3: multiple command line -Z of valid Zipcode records

```

@dagron27 →/workspaces/csci331-spring-24-project4-dagron27 (main) $ ./proj4_norm -Z99950 -Z501 -Z97134
Zip Code: 99950
Place Name: Ketchikan
State: AK
County: Ketchikan Gateway
Latitude: 55.3422
Longitude: -131.648

Zip Code: 501
Place Name: Holtsville
State: NY
County: Suffolk
Latitude: 40.8154
Longitude: -73.0451

Zip Code: 97134
Place Name: Oceanside
State: OR
County: Tillamook
Latitude: 45.4549
Longitude: -123.965

```

Example 4: multiple command line -Z of valid Zipcode records and one invalid. This execution uses a variant with random row organization and reorganized columns.

```

@dagron27 →/workspaces/csci331-spring-24-project4-dagron27 (main) $ ./proj4_rand_state -Z99950 -Z51 -Z97134
Zip Code: 99950
Place Name: Ketchikan
State: AK
County: Ketchikan Gateway
Latitude: 55.3422
Longitude: -131.648

No record found for zipcode: 51

Zip Code: 97134
Place Name: Oceanside
State: OR
County: Tillamook
Latitude: 45.4549
Longitude: -123.965

```


Trouble Shooting

Example 1: Valid input but Zipcode data not found

NOTE: this value is used as an example, it was not meant to be in the data

```
@dagron27 →/workspaces/csci331-spring-24-project4-dagron27 (main) $ ./proj4_norm -Z54
No record found for zipcode: 54
```

```
@dagron27 →/workspaces/csci331-spring-24-project4-dagron27 (main) $ ./proj4_norm
Please enter a zipcode: 54
No record found for zipcode: 54
```

Solution: Check the data set and ensure that the value is in fact present.

Example 2: incorrect input (-Z flag)

```
@dagron27 →/workspaces/csci331-spring-24-project4-dagron27 (main) $ ./proj4_norm 544
```

```
@dagron27 →/workspaces/csci331-spring-24-project4-dagron27 (main) $ ./proj4_norm
Please enter a zipcode: -Z54
No record found for zipcode: 0
```

Solution: -Z is needed with command line and should not be used with direct user input.

Example 3: incorrect user input, multiple entries

```
@dagron27 →/workspaces/csci331-spring-24-project4-dagron27 (main) $ ./proj4_norm
Please enter a zipcode: 99950 501 97134
Zip Code: 99950
Place Name: Ketchikan
State: AK
County: Ketchikan Gateway
Latitude: 55.3422
Longitude: -131.648
```

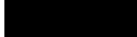
Solution: use -Z for multiple entries. User input will only accept the first entry.

CLOSING REMARKS

Hopefully this user guide has provided you with all the information you need to effectively use our system. We've aimed to make this document comprehensive and easy to understand, covering all the features and functionalities of the system.

Feedback is greatly appreciated moving forward to help improve user experience. Contact me at the provided email address:

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